

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS /R

M2 Practical No: 8

Aim: Performing two-way ANOVA using aov() (R).

Code:

```
library(readr)

df <- read.csv("kepler-exoplanet.csv", stringsAsFactors = TRUE)

colnames(df)[5] <- "Gender"
colnames(df)[6] <- "CGPA_Range"
colnames(df)[8] <- "Depression"

convert_cgpa <- function(x) {
  x <- trimws(as.character(x))
  if (x == "CANDIDATE") return(1.0)
  if (x == "FALSE POSITIVE") return(0.0)
  # Adding CONFIRMED if present in the specific column
  if (x == "CONFIRMED") return(2.0)
  return(NA)
}

df$CGPA_Numeric <- sapply(df$CGPA_Range, convert_cgpa)

model <- aov(CGPA_Numeric ~ Gender * Depression, data = df)

summary(model)

aggregate(CGPA_Numeric ~ Gender + Depression, data = df, FUN = mean)
```

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Output:

The screenshot shows the RStudio interface with the following content:

```
R - R 4.5.2 - ~/R
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
> library(readr)
> df <- read_csv("kepler-exoplanet.csv", stringsAsFactors = TRUE)
> colnames(df)[5] <- "Gender"
> colnames(df)[6] <- "CGPA_Range"
> colnames(df)[8] <- "Depression"
> convert_cgpa <- function(x) {
+   x <- trimws(as.character(x))
+   if (x == "CANDIDATE") return(1.0)
+   if (x == "FALSE POSITIVE") return(0.0)
+   # Adding CONFIRMED if present in the specific column
+   if (x == "CONFIRMED") return(2.0)
+   return(NA)
+ }
> df$CGPA_Numeric <- sapply(df$CGPA_Range, convert_cgpa)
> model <- aov(CGPA_Numeric ~ Gender * Depression, data = df)
> summary(model)

      Df Sum Sq Mean Sq F value Pr(>F)
Gender      2 2338.3   1169.2  470640.4 <2e-16 ***
Depression  1    0.4      0.4    145.9 <2e-16 ***
Gender:Depression  1  20.0    20.0  8054.1 <2e-16 ***
Residuals    9559   23.7     0.0
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> aggregate(CGPA_Numeric ~ Gender + Depression, data = df, FUN = mean)
      Gender Depression  CGPA_Numeric
1  CANDIDATE          0  1.0000000
2  CONFIRMED          0  0.9894366
3 FALSE POSITIVE      0  0.0000000
4  CONFIRMED          1  0.0000000
5 FALSE POSITIVE      1  0.0000000
>

Environment History Connections Tutorial
R - Global Environment
Data
anova_model List of 13
data 212 obs. of 5 variables
data_paired 2930 obs. of 2 variables
df 9564 obs. of 51 variables
Formula.1.const 212 obs. of 5 variables
kepler.exoplanet 9565 obs. of 50 variables
long_data 5860 obs. of 2 variables
meteorite.landi 45716 obs. of 10 variables
model Large aov (13 elements, 941.3 kB)
paired_t_test List of 10
Values
difference int [1:2930] -1656 -896 -1329 -2110 -227 ...
Functions
convert_cgpa function (x)
```

Files Plots Packages Help Viewer Presentation

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